🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 8 · STANDARD 6.DS.5

Display Data: Histograms

Lesson 8-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can make and read a histogram to display data in intervals.
Language Objective: I can explain my histogram using the words histogram, frequency, interval, and distribution.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Histogram **Frequency** **Interval** **Distribution** **Variability**

👆 *Tap any word to see what it means and a picture.*

1. A bar graph that shows how often data falls into equal intervals is a ________.

2. How many data values fall in an interval is the ________.

3. An equal-sized range of values used to group data is an ________.

4. The overall pattern of how data is spread out is the ________.

5. How much the data values differ is the ________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

As you build the frequency table with intervals of 10, what does the height of each bar in the histogram represent?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **Histogram** — A bar graph that groups data into equal ranges. The bars touch.
  Histograma — Una gráfica de barras que agrupa datos en rangos iguales. Las barras se tocan.
- ☐ **Frequency** — How many times a value shows up.
  Frecuencia — Cuántas veces aparece un valor.
- ☐ **Interval** — A range of numbers used to group data.
  Intervalo — Un rango de números para agrupar datos.
- ☐ **Distribution** — How the data is spread out.
  Distribución — Cómo están repartidos los datos.
- ☐ **Variability** — How spread out the numbers are.
  Variabilidad — Qué tan separados están los números.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The height of each bar shows the ____ of that interval.
La altura de cada barra muestra la ____ de ese intervalo.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A histogram groups data into equal intervals so large data sets become readable.

Evidence: Students explain grouping into intervals makes 30 values readable and that each bar's height is the frequency for that interval.

Reasoning: This evidence connects the definition of histogram to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The height of each bar shows the ____ of that interval.**
  La altura de cada barra muestra la ____ de ese intervalo.
- **The tallest bar is the interval where ____.**
  La barra más alta es el intervalo donde ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Histogram
2. **Notes 2:** Frequency
3. **Notes 3:** Interval
4. **Notes 4:** Distribution
5. **Notes 5:** Variability
6. **Watch for this mistake:** *A common mistake in Display Data: Histograms is counting a boundary value in two intervals at once — for example, putting the value 10 in both the '0–9' bin and the '10–19' bin. Each data value belongs to exactly ONE interval. Before you submit, add up all your frequencies and check that the total matches the number of data values you started with.*
7. **Try It 1:** D. 10–19 with 9 players *— The 10–19 interval has the greatest frequency (9 players), so its bar is the tallest.*
8. **Try It 2:** B. 16 *— Add every bar's frequency: 2 + 8 + 5 + 1 = 16 players total.*